



A simple threshold captures the social learning of conventions

Douglas Guilbeault^{a,1,2} , Spencer Caplan^{b,c,1,2} , and Charles Yang^{d,e,1,2}

Affiliations are included on p. 11.

Edited by Thomas L. Griffiths, Princeton University, Princeton, NJ; received April 4, 2025; accepted March 17, 2026, by Editorial Board Member Paul DiMaggio

A persistent puzzle throughout the cognitive and social sciences is how people manage to learn social conventions from the sparse and noisy behavioral data of diverse actors, without explicit instruction. Here, we show that the dominant theories of social learning perform poorly at capturing how individuals learn conventions in coordination experiments that task them with matching their behaviors while interacting in social networks. Across experiments, participants' choices systematically deviate from both imitation and optimization. Instead, they follow a categorical, two-stage learning process: they behave probabilistically until they acquire enough information about each other to trigger a mental threshold and then their choices stabilize. We effectively estimate this threshold using the tolerance principle (TP), a parameter-free equation developed to model how children learn rules in language. We show that threshold-based agents produce social learning that is more accurate than imitating and optimizing agents, while also providing a better model of how a critical mass of dissenters can overturn conventions. The superior performance of our model holds when comparing against a variety of optimization approaches, including Bayesian inference. Furthermore, in a preregistered dyadic experiment requiring people to infer nonlinguistic behavioral patterns amid controlled levels of noise in observed signals, TP outperforms all other models at reproducing learning rates among human participants. These findings offer compelling evidence that a simple, mathematical threshold underlies individual and social learning, from grammatical rules to behavioral conventions.

conventions | social learning | optimization | coordination | social networks

One of the core unsolved questions in the study of human cultural evolution is how social organization emerges—in the form of conventions and norms—from an initial state of widespread behavioral and cognitive variation, with no centralized authority to impose and facilitate coordination. A longstanding though tentative answer to this problem is that social organization emerges spontaneously, without the need for a centralized authority, as a byproduct of diverse agents trying to coordinate in large, well-connected social networks (1–12). However, what this macroscale insight does not address is how individuals manage to learn conventions when observing only a sparse sample of inconsistent actions from diverse actors, with no regularity in collective behavior yet established and no explicit guidance. This feat is especially puzzling since, for a global convention to emerge from an initial state of incoordination, agents often have to select behaviors from a continuously evolving stream of competing alternatives that lack a priori differences in their intrinsic value or fitness. For instance, there is no intrinsic difference in the quality of greeting gestures, such as handshaking or bowing; the primary criterion for convention viability is whether it is sufficiently popular to enable reliable coordination. (2, 6, 8, 9, 11–13). How do people learn which conventions to adopt when collective behavior is in flux and no choice is intrinsically better than any other?

The question of how humans learn conventions from a sociocognitive perspective has been explored over decades by a wide range of qualitative and quantitative approaches (4, 8, 9, 13–16). Regarding efforts to formally stipulate and model the cognitive function for how humans solve this learning problem, two proposed mechanisms of a general nature are paradigmatic, namely: imitation and optimization. The imitation approach makes minimal assumptions by modeling human social learning as driven simply by individuals copying each other's behavior (6, 17–20). Despite providing a reasonable approximation of social learning in various settings (6, 21–24), this imitation mechanism often leaves underspecified the cognitive process underlying how people perceive, detect, and learn patterns of behavior when observing continuously evolving and diverse stimuli;

Significance

Social conventions—such as how people dress and greet in the workplace—pervade human culture. Yet how people learn conventions as they emerge and evolve in social networks is an enduring puzzle. Contrary to dominant approaches, we show that modeling humans as imitators or optimizers fails to capture how people learn conventions across dyadic and network-based coordination experiments. Instead, people only process information about each other until a mental threshold is triggered and then their behaviors stabilize. We effectively estimate this threshold using the tolerance principle, a parameter-free equation developed to explain how children learn rules in language. This shows how the social learning of conventions may be enabled by simple, categorical mechanisms.

This is a preprint.

Author contributions: D.G., S.C., and C.Y. designed research; performed research; analyzed data; and wrote the paper.

The authors declare no competing interest.

This article is a PNAS Direct Submission. T.L.G. is a guest editor invited by the Editorial Board.

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¹D.G., S.C., and C.Y. contributed equally to this work.

²To whom correspondence may be addressed. Email: dguilb@stanford.edu, scaplan@gc.cuny.edu, or ycharles@upenn.edu.

This article contains supporting information online at <https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.2508061123/-/DCSupplemental>.

Published April 22, 2026.

for example, as we explore in detail below, formal models of imitation in the context of conventions often assume agents do not possess or learn from their memory, which significantly limits the ability to posit or describe cognitive learning functions (8, 10).

The second approach—optimization—seeks to address this issue by framing social learning as a prediction problem. On this view, people make extensive use of information about each other's past and present behaviors to predict which action is most likely to match the behaviors of others. This account frequently suggests that humans are focused on maximizing the accuracy of their beliefs through statistical reasoning (10, 25–29), while also seeking to develop conventions that optimize the efficiency and speed of communication (30–33). The apex of this approach is, perhaps, the growing use of large language models (LLMs) to simulate humans; these models are designed to leverage massive amounts of data (e.g., all language data on the internet) using expensive, brute force computations for the purpose of optimally predicting behavioral outcomes, such as people's word choices in everyday communication (34–36). In recent years, the view of human coordination as driven toward statistically optimized communication has become the dominant approach to modeling social learning dynamics, both in general and in the context of conventions specifically (10, 28, 33, 37–43).

Yet, the optimization view is at odds with decades of work in organizational behavior (44, 45) and cognitive science (46, 47) showing that humans are satisficers (48)—i.e., agents who only gather and process a limited amount of information to arrive at “good enough” (satisfactory) beliefs and behaviors. Core to the theory of satisficing is the concept of a threshold: the idea that people only gather information up to a point—sometimes defined as an “aspiration value” (45, 49)—after which their behaviors stabilize. Numerous studies show how collective behavior is effectively modeled by assuming individuals learn via thresholds (50–54). However, threshold models often struggle to formally identify and empirically estimate the threshold underlying people's adoption of behavior. Various attempts at modeling satisficing thresholds have been made, but they often assume many hyperparameters that are context-specific (49, 55–58) and independent of psychological theory (45), such as industry-specific criteria for “good enough” performance in organizations (59, 60). The proliferation of such models is the reason why one of the most authoritative textbooks on artificial intelligence states that while satisficing “appears to be a useful model of human behavior in many cases [...] it is not a formal specification for intelligent agents, however, because the definition of good enough is not given by the theory” (61). The lack of an empirically grounded, context-free definition of threshold learning contributes to why threshold models have gained limited traction in accounts of social conventions.

In this study, we develop a general threshold model that estimates how much information about each other's behavior is good enough for people to select a stable convention for coordination. Our model is parameter-free, context-general, and does not assume a theory-laden value parameter as is commonly evoked in utility functions for optimizing agents. Instead, our framework identifies a deterministic threshold that makes precise predictions for how much information is needed for people to learn any rule-like regularity from noisy data involving frequent violations of the rule. To define this model, we use the concept of the tolerance principle (TP), which was originally developed to identify how much data is good enough to allow children to learn grammatical rules in a threshold-based fashion from noisy and sparse input with many grammatical exceptions (62). In the

next section, we introduce the TP, its grounding in research on language acquisition, and we summarize evidence supporting the expectation that TP captures rule-learning across domains, including arbitrary patterns, arithmetic rules, and—as is the contribution of this study—behavioral conventions. We then test the ability for TP to estimate how people learn conventions across several experimental paradigms. First, we compare choice models—optimization, imitation, probability matching, and TP—in their ability to predict human decisions in coordination experiments involving the “name game” (6, 10), which captures the emergence of conventions and their disruption via dissenters. We then report the results of a preregistered experiment designed to compare these models in their ability to predict how people learn nonlinguistic coordination choices under different levels of noise in available behavioral signals.

The TP. The TP is a mathematical threshold originally designed for rule formation in language (62). Linguistic rules almost always have exceptions: The English past tense rule “add -ed” does not apply to the irregular verbs but all speakers learn the rule as young children and can readily apply to new verbs such as “google” (“googled”) when it became a verb in the 1990s. It is recognized across a wide spectrum of theories (63–66) that a rule must be supported by a critical mass of evidence to overcome the exceptions. Under various assumptions about the statistical distribution of words, it is possible to prove that a pattern defined over N unique items can only generalize if the number of exceptions does not exceed the threshold of $N / \ln N$: otherwise, it is more efficient to memorize all items by rote without positing a generalizable rule (62). The TP is a descriptive theory of how individuals make decisions based on their subjective experiences of the environment, as in the case of a child language learner's growing vocabulary of words (i.e., N in TP). As such, decisions are always provisional: a rule can be applied productively as long as it is good enough to cover a sufficiently high proportion of N .

Although originally formulated as a learning theory for language (and demonstrated across language families), experimental evidence suggests that the TP may function as a general regularity detector in implicit learning across domains, which tracks ratios via the statistical learning of transitional probabilities (67). For example, experimental evidence (68) shows that infants can generalize an arbitrary pattern if 11 out of 16 training items follow the pattern but not 10 out of 16, which—despite covering a dominant majority that ought to generalize under an optimization approach—has six exceptions, falling short the TP threshold of 5 (16/ $\ln 16$). Related studies suggest that TP performs well at modeling when children learn generalized numerical relations (such as the ability to count indefinitely) during the acquisition of mathematical knowledge (69). The TP is thus a candidate theory of how people identify good enough generalizations in domains not limited to language, and the similarities between rule formation in language and social behavior provide additional impetus for applying the TP to behavioral conventions (4).

The Name Game Experiments. To test whether the TP provides an accurate model of convention learning and formation, we examine the dynamics of the name game. The name game is a coordination game (3, 5, 6, 10) in which players are tasked with agreeing on a shared name for a human face while interacting in structured social networks. On each round, each player is randomly assigned to attempt to coordinate with one other person in their network (Fig. 1). Each player independently

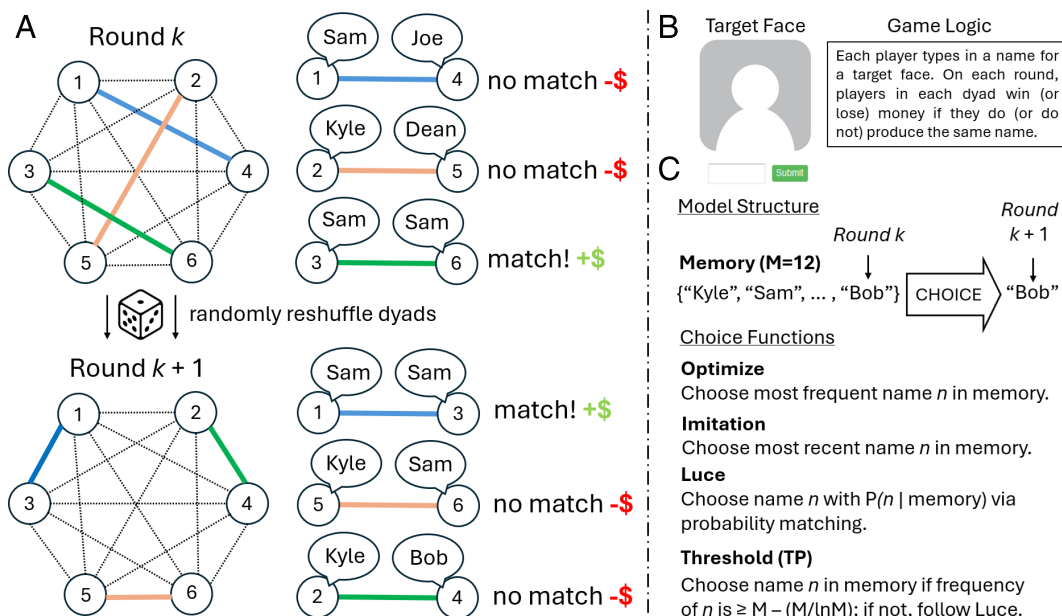


Fig. 1. This schematic displays the logic of the name game and the agent-based models (ABMs) used to simulate and predict participants' round-by-round choices. Panel (A) shows an example fully connected network of six nodes (the empirical networks we study are larger, including 24, 48, or 96 nodes) and illustrates that, on each round, participants are randomly paired with one other person and asked to independently type in whichever name they want for a target face, shown in panel B. Panel (C) displays the basic structure of agents in each ABM and briefly outlines how each of the ABMs differ in their choice function for sampling an agent's choice on round $k + 1$ based on their sequential memory of past names encountered on previous rounds. Our main simulations assume the size of memory (M) is 12. Note: Please see *The Models* for full and accurate description of how each model was precisely implemented; this figure provides the gist.

provides a name, and they are financially rewarded (punished) if they provide an identical (different) name. The arbitrariness of naming an unknown face makes the name game a direct test case for the core learning problem in convention formation, since all names can, in principle, work for solving this task. The payoff for each name depends on whether other agents use this name. ABMs and experiments with human participants show that, in the early stages of the name game, many names (possible conventions) are proposed, and there is instability in which names repeat across interactions (6). The puzzle is how agents can learn and converge on a particular name that coordinates with other players, while beginning in a state of social discord and convention diversity.

A striking result in a recent experiment with human participants playing the name game is that, in fully connected networks—in which all players can be randomly paired with anyone else—a symmetry-breaking event occurs after an extended period of unstable coordination, such that a single name sharply rises as the globally dominant option, rapidly spreading throughout the network (6). The study finds that regardless of the size of fully connected networks, conventionalization takes place in about 20 to 30 rounds; this was replicated over six trial networks, three of size 24, two of size 48, and one of size 96, totaling 264 participants.

A follow-up study (10) investigates how established conventions in the name game may be broken and changed. After fully connected networks of human players converged on a shared name, the experimenters surreptitiously introduced a committed minority of confederates who always promote an alternative name. Across 10 trials, Centola et al. (10) show that if the size of the committed minority reaches a tipping point (averaging 25% of the network), the previously established convention flips; but so long as the committed minority remains below the tipping point, the convention remains intact.

These results from the name game provide important insights into the problem of conventionalization. However, they primarily concern macrolevel outcomes and do not investigate the individual-level learning mechanisms that explain how

individuals come to identify and converge upon a shared name within a cacophony of competing conventions, and how they come to break away from it. In our study, we focus on modeling experimental data of the name game in the fully connected networks examined and published by Centola and Baronchelli (6) as well as Centola et al. (10). Each ABM under comparison provides a different candidate specification of the underlying mental process by which agents make their coordination choice on the next round by sampling from their memory of past interactions in the name game. We describe each model in detail below.

The Models. In all models, the behaviors that an agent encounters via previous interactions are stored in a first-in-first-out queue that can hold a finite number (M) of entries, following the “modal” model of memory (70). Each of the N agents starts with a random name chosen from a Zipfian distribution, as shown in the empirical data of the experiments, and pushes it into their memory queue. During each round of interaction, two agents are paired randomly as in a fully connected network, and each produces a name according to the procedure described below (Fig. 1). If the paired agents produce the same name, then that name enters the queue, squeezing out the least recent entry if the memory size M has been reached. If the paired agents produce different names, then they each push the other's name into the queue and eject the least recent entry that contains the name they themselves produced.

Holding this memory model constant (see *SI Appendix, Tables S1–S6* for fully enumerated examples of each model's choice function), we compare four different approaches to simulating how agents draw coordination options from memory when deciding which convention to attempt on a subsequent round (Fig. 1C). We compare: i) the imitation (IM) model [first developed by Baronchelli et al. (5) and extended by Centola and Baronchelli (6)], where agents choose names randomly from memory, but when an agent encounters a name previously stored in their memory, they delete their entire memory and only retain the successful “repeat” name; ii) the optimization (OP) model

[proposed by Centola et al. (10)], which follows the dominant approach in social learning by assuming agents always select the most frequent choice in their memory (and randomly when faced with ties), and memory of past interactions is not deleted after coordination success (random sampling with ties is applied in all simulations); iii) the Luce model [based on the Luce Choice axiom; (71)], whereby agents choose names from their memory at a probability that is directly proportional to the frequency of each name in memory (i.e., probability matching); and iv) the threshold learning (TP) model, whereby agents learn conventions according to TP. As an example of TP learning, for $M = 12$ (i.e., 12 slots in the memory queue), if one choice occupies at least $M - M/\ln M$ (where $M/\ln M$ represents the maximum number of exceptions a rule can tolerate) or 8 entries in the queue, then the choice becomes “good enough” as it is sufficiently dominant over the alternatives, and the agent always provides that choice (i.e., it becomes “productive”), much like how a grammatical rule is supported by a sufficient number of rule-abiding words over the exceptions (62). That is, the alternative names that are not yet conventionalized (i.e., not yet “rule-following”) are analogous to exceptions. If none of the choices is productive in this sense, then under the TP model the agent samples a choice randomly from the queue following Luce’s Choice Axiom.* Note: The OP model can be considered a pure optimization model in the economic sense. It specifies a learning process that maximally exploits information to predict and select the most likely expected outcomes for the purpose of accelerating the emergence of a coordination system that fulfills as many people’s preferences as possible (where the assumption is that everyone in the name game prefers to coordinate) (72).

These models are compared on two aspects of the findings with data from refs. 6 and 10. First, because the round-by-round choices and interactions are recorded for all participants, we can reconstruct and simulate the history of participants’ memories. The IM, OP, and TP models all make deterministic predictions under certain conditions: IM predicts that a match will always be reproduced in the next round; OP predicts that the most frequent choice in the memory (excluding ties) will supply the next round choice; and TP makes a categorical prediction that a choice will only become productive (i.e., stably chosen on the subsequent round) once the TP satisficing threshold is met. Second, we use each model to simulate the entire dynamics of the name game within an ABM framework, and compare how well the evolutionary dynamics (e.g., the global convergence rate) in each ABM correspond to the real-time evolutionary dynamics observed in networks of human participants, in terms of both the emergence and change of conventions.

Results from the Name Game

Evidence of Threshold Learning. We begin by testing whether participants’ raw round-by-round behavioral data in the name game exhibits the two-stage threshold dynamics predicted by TP. For each participant, we measure how their coordination choices change based on whether a name is likely to have reached

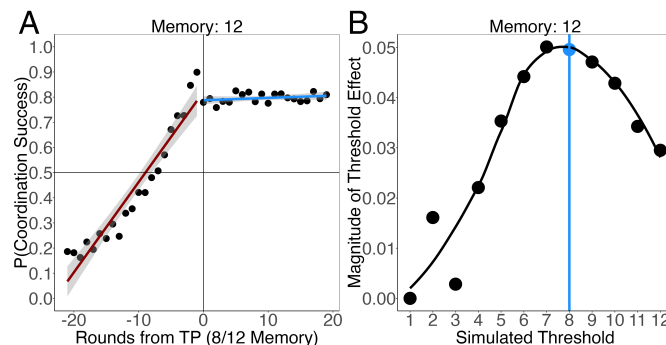


Fig. 2. Measuring individual participants’ round-by-round choice of convention relative to whether the TP threshold has been reached in their simulated memory. To simulate agents’ memory, we track the history of empirical interactions encountered by each agent in each network trial from Centola and Baronchelli (6) and Centola et al. (10), and we assume that agents sample their next choice from M (memory) past interactions. Our main results assume that M is 12, consistent with prior analyses of the name game (10); but all results are consistent across a wide range of plausible M values. Panel (A) shows how the rate of individuals’ coordination success in convention choice stabilizes across participants once TP is reached (suggesting convergence in choice of convention). Separate OLS regression lines are plotted before and after TP is reached. Red (blue) lines show a standard ordinary least squares curve fit for all behavioral data before (after) a name reaches TP in agents’ memory; data are averaged across all agents and trials. Error bands show 95% CIs. Panel (B) replicates panel (A) while assuming the full range of possible values for agents’ threshold and plots the magnitude of the regression discontinuity effect (interaction of round number with the center point: comparing before and after the threshold has been reached) for each simulated threshold. The actual TP threshold is indicated in blue.

the threshold predicted by TP in the participant’s memory. To simulate participants’ memory, we track the history of empirical interactions encountered by each participant in each network, and we assume that participants are sampling their next choice of name from M (memory) slots. Our main results assume that M is 12, consistent with prior analyses of the name game (10); but all results are consistent across a wide range of M values (SI Appendix, Figs. S1 and S2) and also when randomly varying M across agents in the same network (SI Appendix, Fig. S3).

Fig. 2 compares participants’ coordination choices before and after a name satisfies TP in their simulated memory (results are averaged across all participants in all trials of fully connected networks (6, 10) excluding any confederate data). Panel A displays the probability that a participant succeeds in coordinating with their round partner as a function of reaching TP. We find that early in the game, when participants are still 10 to 20 rounds away from reaching TP in their memory, the rates of coordination success are low (significantly lower than 50%; $P < 0.0001$, Student t test, Two-tailed). Yet, as the frequency of a name in participants’ memory approaches TP, the rate of coordination success (i.e., the rate at which agents choose the same name) sharply increases. Critically, as soon as TP is reached, there is a qualitative shift in the rate of social learning consistent with a two-stage learning process. After TP is satisfied, the rate of coordination success plateaus at approximately 80%. Regression discontinuity analysis indicates a significant reduction in learning rate before and after TP is reached ($P < 0.00001$, Student t test, Two-tailed). In other words, once TP is satisfied, participants converge on a shared name, and they no longer change which name they use, such that social information from subsequent rounds has little to no impact on their behavior. The threshold effect captured by Fig. 2 does not merely reflect the steady emergence of a global convention in the network synchronizing players’ responses, since there is notable variation

*Since the TP threshold ($N/\ln N$) uses the natural logarithm, it will inevitably produce a real number, whereas the number of items in memory, or the number of words used in rule learning, is always an integer, since words are considered to be discrete units. For example, for $N = 16$, $16/\ln 16 = 5.77$. While this value rounds up to 6, having 6 exceptions exceeds the threshold ($6 > 5.77$) and should be qualitatively treated as doing so. However, as we discuss throughout our results section, all of our findings are robust to including variation in agents’ memory size and noise in agents’ implementation of TP, which demonstrates that our findings are not sensitive to this flooring mechanism and do not require that TP represents the same threshold across individuals.

in when individual players reach TP within each network trial (SI Appendix, Fig. S4).

To further probe this point, we evaluate the extent to which optimization predicts participants' behavior as a function of whether a name has reached TP in participants' memory. When participants are 10 to 20 rounds before reaching TP, the optimizer model performs strikingly poorly at predicting participants' next round choice (achieving an accuracy rate of only 31% on average). As the game proceeds, the optimizer model steadily improves, which at first seems to suggest that participants increasingly exhibit optimization behavior as they continue to learn from their social experiences. However, the ability for optimization to predict participant behavior suddenly plateaus precisely when a name reaches TP in participants' memory, which is a two-stage learning process not predicted by the OP model (SI Appendix, Fig. S5).

Having demonstrated evidence of threshold dynamics in participants' learning behavior, a natural next question is whether TP is an especially good fit for capturing this threshold. To evaluate this question, we reproduce the analysis underlying Fig. 2A across the full range of possible thresholds, while holding M constant at 12. We then calculate the magnitude of the threshold effect separately for each simulated threshold by replicating the regression discontinuity technique of quantifying the statistical significance of the change in slope before and after the threshold is reached (73).[†] The distribution of threshold effects across all possible thresholds is shown in Fig. 2B (assuming $M = 12$). The TP identifies a threshold (8/12) that is at the peak in terms of capturing a qualitative shift in participants' coordination behavior, neck-in-neck with its neighbor value of 7/12. That 7/12 achieves a threshold magnitude slightly higher than the 8/12 threshold identified by the strictly deterministic TP function assuming $M = 12$ is still consistent with the two-stage learning model predicted by TP, since the success of TP does not hinge on the 8/12 being exactly the right cutoff across individuals. Individuals' threshold activation can vary as a function of several unobserved factors, including variation in memory size and behavioral noise (e.g., due to fluctuations of attention during the task); these factors can shift when TP is activated for individuals given the same history of interactions in the task. As we discuss in the "Objections" subsection, TP consistently provides a superior estimation of the underlying threshold effect even when i) varying the assumed memory sizes of agents (SI Appendix, Fig. S1), ii) simulating noise in individuals' threshold activations (SI Appendix, Table S7), and iii) comparing against other competing threshold models such as a 2/3rds rule (SI Appendix, Fig. S6).

Predicting Conventionalization. This evidence of a two-stage, threshold-based learning process suggests that TP should out-compete standard models for predicting individual participants' round-by-round choices of convention in the name game. Here, we directly compare models in terms of their ability to predict participants' round-by-round choices in experimental trials of the name game from Centola and Baronchelli (6) and Centola et al. (10) (data from the latter excludes any behavioral data from confederates in the task). For this analysis, we compare each model's ability to predict participants' choice of convention on

the next round using a simulated memory M populated with their actual past interactions in the experimental fully connected networks.

As Fig. 3A shows, across a range of plausible memory sizes (M), the TP model achieves notably higher accuracy than all other models. Panels B and C zoom in on these results in the $M = 12$ case. The two-stage learning process theorized by TP predicts that, prior to TP being reached, people will sample according to a probability matching model (i.e., the Luce axiom). Fig. 3B confirms that, during the first 12 rounds of the name game prior to the saturation of people's memory, the frequency of a name in participants' memory is highly correlated with the probability of that name being chosen ($r = 0.84$, $P < 2.2 \times 10^{-16}$, Pearson Correlation, Two-sided), as predicted by Luce sampling. These findings alone limit the plausibility of the OP model, which predicts that participants will choose the plurality (most frequent) name in memory 100% of the time. However, during this period, while the plurality prethreshold name occupies 48% of total memory slots on average, it is chosen by participants only 61% of the time (just moderately above a coin flip), rather than deterministically (100% of the time) as predicted by OP. The Luce model does a better job than optimization at capturing the dynamics of participants' round-by-round choices in the early stages of the game. SI Appendix, Fig. S7 confirms the robustness of this correlation across a range of sensitivity tests.

Yet, after participants' memory is saturated, their choice behavior falls out of line with Luce and becomes much more effectively predicted by the threshold model, as predicted by the two-stage TP process. Fig. 3C compares all models in their ability to predict participants' round-by-round choice of convention based on the simulated memory histories after 12 rounds. From this point on, the TP model provides a significantly more accurate prediction of participants' next choice of convention, regardless of the round. Specifically, Fig. 3C shows that the TP model accurately predicts participants' choices 87.9% of the time, which is significantly higher than the imitation model (81.1%) the optimizer model (75.2%), and the Luce model (53.4%), which performs poorly after the initial stages of the experiment (for all comparisons: Proportion Test, $P < 0.00001$, Two-sample, $n = 8,893$ trial-level observations). TP also significantly outperforms alternative threshold specifications, such as a 2/3rds rule (SI Appendix, section A.7).

Surprisingly, the optimizer model performs worse than the simple imitator model in this case. The performance rates of the optimizer model further decrease if we only examine its predictions before a simulated name reaches TP in participants' memory, yielding an accuracy rate of only 59.1%, which is significantly lower than both the imitator and the threshold (TP) model (for both comparisons: Proportion Test, $P < 0.00001$, Two-sample, $n = 10,630$ trial-level observations).

Moreover, despite its decent performance in this setting, a detailed analysis of participants' memory updating casts doubt on the plausibility of IM. Consider IM's memory procedure: if, as that model predicts, participants delete their memory after successfully encountering a name twice, then the odds of returning to a previously produced name following an eventual switch should be no higher than the rate of production on the first round (i.e., from the initial, highly diverse, distribution of produced names). We find, however, that when participants switch names between adjacent rounds (a total of 5,029 trials across all networks) then $P(\text{"said before"} | \text{switched}) \approx 51\%$; that is, they provide a name from past history in such cases a majority of the time, clearly indicating a role of memory.

[†]Specifically, we ran a standard OLS regression that uses the "rounds from TP" variable on the horizontal axis of Fig. 2 to predict "the probability of coordination success" variable on the vertical axis, while including a dummy variable to indicate, at the individual-level, whether the corresponding round is before or after TP being reached by the individual, along with an interaction term between this dummy variable and individuals' round distance from TP. This interaction term measures whether the change in slope before and after TP is reached marks a significant difference, and the size of coefficient of this interaction term indicates the magnitude of this threshold effect. See ref. 73 for more information on the motivating methodology.

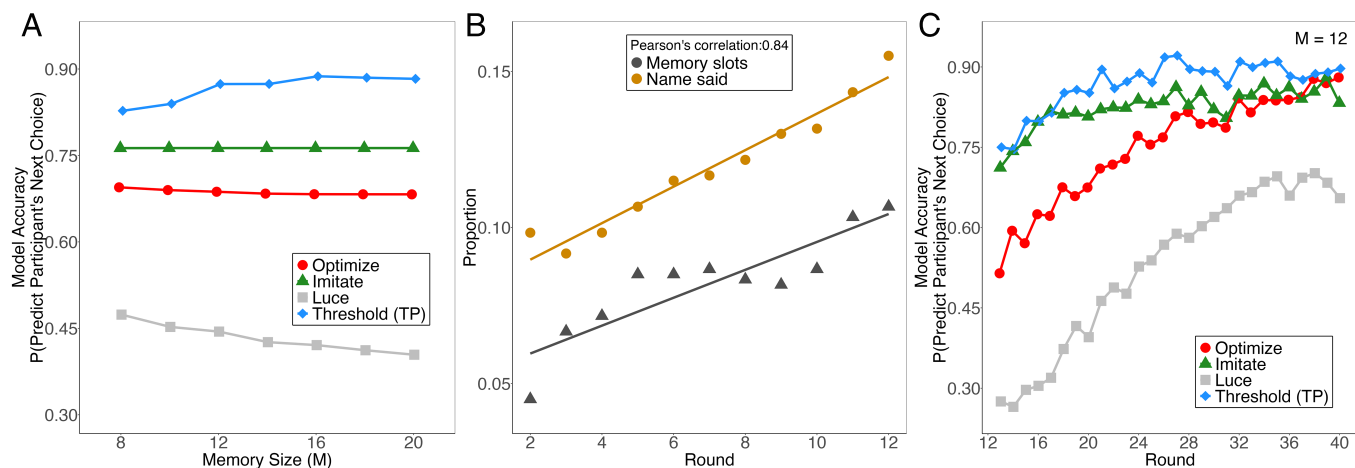


Fig. 3. Comparing ABMs in how accurately they predict the next round choices of human participants in experimental data from fully connected social networks in the name game (6, 10). To simulate participants' memory, we track the history of empirical interactions encountered by each participant in each network trial, and we assume they sample their next choice from M (memory) past interactions. Panel (A) shows the round-by-round accuracy of each ABM across M values. Panel (B) evaluates the fit of the Luce axiom for participants' choices in the initial rounds of the game prior to their memories being filled (assuming $M = 12$); it shows the correlation between the frequency of a name in participants' memories and the probability of that name being chosen by participants on the next round. Panel (C) shows the round-by-round accuracy of each ABM's predictions of the experimental data for the later stages of the experiment after memories are full ($M = 12$). In (C) all models except for Luce make deterministic predictions (except for infrequent ties in OP); but the Luce model is stochastic, and the data present the average accuracy rate of Luce across 24,464 round level observations, with an average of 269 observations per round across networks (results are robust across replicated simulations; *SI Appendix, Fig. S2*).

Finally, we note that only the TP model offers a good fit to the empirical rate of convergence to a shared convention across entire networks. Centola and Baronchelli (6) found that in fully connected networks, the number of agents ($N = 24, 48, 96$) had little influence on the number of rounds required for social networks to converge on a globally shared name, which lies in the range of 20 to 30 rounds. The convergence time for TP is highly consistent with this experimental finding across network sizes, while both IM and OP converge far too quickly (*SI Appendix, section A.10 and Table S8*). Luce, by contrast, converges too slowly, and yields different convergence rates across network sizes.

Predicting Convention Change. These findings provide evidence that a two-stage threshold-based model is effective at predicting how people learn and form conventions from scratch in experimental data on coordination in fully connected social networks. We proceed to test whether TP provides an accurate model of when established conventions change. In their experiments, Centola et al. (10) first allowed fully connected networks of human participants to reach consensus on a globally shared name, and then they introduced a critical mass of confederates who coordinated in the attempt to flip the established convention to a specific alternative (all confederates in a network pushed the same alternative). Experimental nonconfederates were oblivious to the injection of confederates, who they perceived to be other players. These experimental data provide an especially conservative test of our theory, since after a particular convention is established, only two names are competing in the social network: the established convention and its proposed alternative by the confederates. This simplification of the dynamic allows us to precisely measure whether—and if so, how—human participants follow threshold-like behavior at the individual-level when switching from the established (rule-abiding) to the alternative (rule-breaking) convention. For this analysis, we first initialized all agents with a full memory ($M = 12$) of the same established convention, and then agents were exposed to a critical mass of confederates pushing the same alternative convention. The data show 3,000 independent simulations of each model

tuned to the same network and critical mass size examined in the five empirical trials from Centola et al. (10) in which a critical mass successfully induced convention change.

For each empirical trial, we identify the earliest round at which the maximum fraction of nonconfederate participants adopted the confederates' name (not all empirical trials exhibited full adoption of the alternative within the observation window despite successful convention change); then, for each simulation of each model, we identify the round at which the same fraction of simulated nonconfederates adopted the confederates' alternative, and we calculate each models' accuracy as the absolute difference in rounds at which the empirical and simulated networks reached this maximum level of convention change. Fig. 4A shows the results of this analysis. TP reproduces convergence rates that are significantly closer to the corresponding empirical trials compared to IM and OP ($P < 0.0001$, Student t test, Two-tailed). Indeed, only TP yields convergence rates that land within the target empirical range for each network size. While the Luce model lands within the empirical range for $N = 24$ networks, it falls outside this range for $N = 48$ and $N = 96$, which indicates inaccuracy not only at the level of each network size, but also across networks, since the empirical networks yielded stable convergence rates regardless of their size. These results are robust to a number of sensitivity tests, including: i) varying agents' memory model (*SI Appendix, Fig. S13*), and ii) varying the number of rounds over which each ABM is observed (*SI Appendix, Fig. S8*).

Fig. 4B evaluates the tipping points identified in each model across a range of network sizes ($N = 24, 48$, and 96). Both the imitation and the Luce model lead to rapid convention change resulting from implausibly small critical mass sizes; even when confederates comprise 10% or less of the population, this is sufficient for triggering convention change under imitation (10). This is inconsistent with the empirical data; in their experiments, Centola et al. (10) report that a critical mass of $\approx 25\%$ is the best fit for their network outcomes with human participants. Thus, while IM performs moderately well at capturing individuals' learning of conventions as they initially emerge (Fig. 3), it performs poorly at accurately reproducing critical mass dynamics in this same

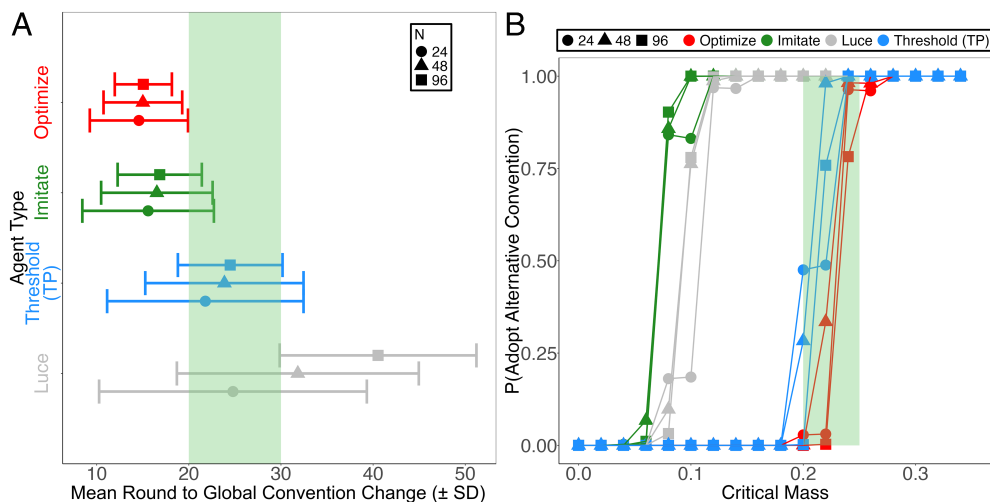


Fig. 4. Comparing ABMs in how accurately they reproduce tipping point dynamics in empirical data on humans playing the name game in experimentally created networks. All data show 3,000 simulations of each model under each parameter configuration. Every agent is initialized with a full memory ($M = 12$) of the same established convention prior to interacting with a critical mass of confederates who push the same alternative name. (A) Comparing ABMs in their accuracy when predicting when human networks converge on the confederates' alternative name (all trials from Centola et al. (10) in which a sufficiently large critical mass is reached). Each ABM for each trial is tuned to the same size of network and critical mass. The green shaded area shows the target range of actual "mean round to global convention change" outcomes for the empirical data. Error bars display one SD of the simulation results. (B) Each ABMs' tipping point behavior across network sizes ($N = 24, 48$, and 96). The vertical axis shows the final proportion of nonconfederate agents (after 75 rounds of interaction) adopting the confederates' alternative name. The horizontal axis shows the proportion of the population filled by confederates. The green shaded area shows the target empirical range of 20 to 25% (10).

setting. Imitation therefore fails as a consistent and accurate model of human learning behavior in the name game that can account for both the emergence and change of conventions.

By contrast, both TP and OP reproduce tipping points as a function of critical mass sizes that are consistent with the empirical data presented by Centola et al. (10). It is worth noting, however, that while OP performs well in this context, OP underperforms at predicting participants round-by-round choices (Fig. 3), and OP converges too quickly in modeling the initial emergence of conventions (i.e., in the absence of confederates) compared to the empirical data (SI Appendix, section A.10). Like IM, OP also fails to provide a robust model of participants' learning behavior in the name game that can capture both the emergence and change of conventions.

The superior performance of TP is reinforced by comparing these models in their ability to predict participants' round-by-round choices in response to confederate influence (10). Here, TP makes a clear prediction: earlier probabilistic adoption of the alternative convention. Consider the case where agents possess a memory with 12 slots. When the original convention fills 8 of these slots and the alternative only 4, both TP and OP predict that agents will always produce the original convention. Yet, when the alternative name reaches 5 slots (decreasing the original convention to 7 slots), TP predicts that agents will start sampling probabilistically since the original convention no longer satisfies TP (which requires 8 slots), whereas OP predicts that agents should continue to produce the original convention, which still occupies the majority of memory.

We test this hypothesis empirically by identifying all cases in all trials from Centola et al. (10)'s experiments in which—consistent with our simulations where $M = 12$ —participants had exactly seven instances of the original convention and five instances of the alternative convention in their memory, independent of the actual round on which this configuration was observed. This yielded a total of 949 observations across all ten networks (regardless of whether the whole network eventually flipped to the alternative). All agents under these conditions produced either the

original or the alternative convention in the subsequent round, providing a clear binary outcome variable for comparing the predictions of the OP and TP model. The OP model predicts that no participants will produce the alternative convention in this case, whereas TP predicts (in accord with Luce's choice axiom) that participants should produce the alternative convention $\frac{5}{12}$, or 42%, of the time. The empirical data show that participants produced the alternative name on the next round in 29% of cases. This outcome is significantly closer to the rate predicted by TP than OP ($P < 0.00001$, Student t test). This is consistent with the tight correlation across the board between the proportion of memory slots that a name occupies (below the TP threshold) and rates of that name's use (SI Appendix, Fig. S7).

These findings are reinforced via a round-by-round behavioral analysis for participants' next-round choices. Using our simulated memory approach (assuming $M = 12$) to examine the time series adoption data (from the initial emergence of conventions to its change) from this tipping point experiment (10), we observe that TP predicts participants' next round choice with 89% accuracy, whereas IM, OP, and Luce achieve 79%, 77%, and 53.9% accuracy, respectively, marking significantly lower performance than TP ($P < 0.0001$ in all comparisons, Proportion Test, Two-sample). The accuracy rates of OP are even lower when only examining cases prior to a name reaching TP. Before TP is reached, OP only predicts participants' behavior under tipping point dynamics with 60% accuracy.

Altogether, these findings indicate that TP provides a robust and accurate model of participants' learning behavior across both the emergence and change (via tipping points) of conventions in the name game, whereas OP and IM are only able to (less accurately) capture parts of these dynamics, while falling short at capturing others. The Luce model generally fails across the board at capturing both convention formation and change. TP stands out in its ability to provide a unifying model of learning behavior that generalizes across the established stages and dynamics in the name game.

Objections. One possible objection is that human behavior is noisy, such that adding noise to the ABMs will improve their performance. However, supplementary analyses show that adding noise systematically worsens model performance, including OP (*SI Appendix*, Figs. S9 and S10) and Luce (*SI Appendix*, Fig. S11). A related objection is that while not all people behave optimally, some more “rational” actors do, and OP may perform better if only a fraction of agents are optimizers and the rest follow Luce (27, 74). However, supplementary analyses show that this approach does not improve OP’s performance (*SI Appendix*, Fig. S12).

We further show that our findings are robust to varying ABMs’ underlying memory model (e.g., using a “bag” model of memory; *SI Appendix*, Figs. S13 and S14). A related concern is whether our results can account for a recency bias in memory (75). On this point, the imitation model is a pure recency model: agents only sample from their most recent memory of a successful coordination. That the imitation model performs poorly at capturing convention learning in this setting suggests that recency bias is unlikely to improve models.

The reader may be further concerned that OP is an overly naive model of statistical optimization, which is better captured by more sophisticated models such as Bayesian inference (25–28, 33, 39, 76, 77). However, supplementary analyses (*SI Appendix*, Fig. S15) show that a standard Bayesian model of rule-learning in language (78, 79) performs poorly at capturing the convergence and tipping point dynamics in the name game. There is no combination of hyperparameter settings in this Bayesian model that enables it to perform well across experimental conditions. Moreover, this Bayesian model fails to reproduce the two-stage threshold learning dynamics captured by TP. It is unlikely that standard Bayesian models provide an effective alternative to OP in this setting. A productive direction for future research is to explore the possibility of recovering threshold learning through extensions of Bayesian frameworks.

Another important objection is that individuals may not all follow TP with the same level of precision. Supplementary analyses show that our findings hold when modeling individuals as following a threshold defined by $TP \pm$ Gaussian noise (*SI Appendix*, Table S7). In this way, our findings are not dependent on TP providing an exactly correct, deterministic description of the underlying threshold. Indeed, other threshold functions may identify similar behavioral dynamics, such as a 2/3rds rule (whereby agents converge on the name occupying 2/3rds of their memory), which identifies an identical threshold as TP when $M = 12$. However, for an alternative threshold to outcompete TP, it needs to achieve higher accuracy rates across different memory sizes, since individuals’ memory capacity varies. Supplementary analyses compare the round-by-round prediction accuracy of TP and a 2/3rds threshold rule under those memory conditions for which each model makes divergent predictions (i.e., $M \in [10, 11, 13, 14]$). In all cases, TP provides significantly more accurate predictions of participants’ coordination choices, capturing on average 22.5 percentage points more of participants’ next round choices than the 2/3rds rule ($P < 0.0001$, Two-Sample Proportion Test; *SI Appendix*, Fig. S6).

While our findings do not rule out alternative thresholds performing well in this setting, our analyses consistently support TP as the strongest and simplest specification of this underlying threshold. These findings are consistent with studies of rule learning across many languages that provide the original motivation for TP (62). That said, ultimately our aim in this study is to demonstrate that a two-stage threshold-based learning

process effectively captures the social learning of conventions, for which TP is a compelling candidate specification for this threshold. More generally, our findings should be taken as support for a threshold-based learning process that TP effectively approximates, rather than as conclusive evidence that TP is the exact description of the underlying threshold in all domains. Verifying the functional form of the threshold at play will require future research with greater access to measurements of microlevel psychological processing, possibly through the inclusion of neuro-imaging data.

The Mind-Reading Game

Nevertheless, a valid concern remains unresolved: our empirical evidence for TP in social learning is based on post hoc analyses of experimental data. Properly testing TP requires an experiment designed for this purpose. To address this, we conducted a preregistered experiment called “the mind-reading game.” In this game, participants are told they are coordinating in a dyadic task; in reality, their interaction partner’s choices followed prespecified sequences, which allowed us to control the level of variability in the partner’s behavioral signals. From the participants’ perspective, they are tasked with identifying and matching their partner’s preferred coordination choice. Specifically, players were tasked with coordinating their choice of binary signals (i.e., choosing the color “red” or “blue”) across a sequence of rounds. Each player is told they are randomized to play as either a “sender” or a “receiver,” where the sender begins by signaling their binary choice, and the receiver then uses the observed signal from the previous rounds to predict the sender’s next choice. Unbeknownst to them, all participants were assigned to play the role of receiver (see *SI Appendix*, Fig. S16 for a schematic overview).

To introduce variation in the strength of the underlying pattern (i.e., the observability of the sender’s preference), participants were told that a random filter occasionally scrambles the sender’s signal, such that the receiver is sent the wrong signal. The participants were told that the scrambler will only be active “on some of the rounds,” but they were not told how often or on which rounds. This created a scenario in which success or failure on a given round could be due to the scrambler. This prompted participants to try to infer the sender’s underlying strategy across rounds, generalizing beyond the noisy feedback on any given round. Under the hood, there was no scrambler. Instead, the stability of the sender’s behavioral signal (i.e., the rate at which the target option, red, was chosen relative to blue) was tuned differently in each condition to derive different predictions from each decision model in terms of the probability that a given agent converges on a stable belief regarding the sender’s preferred choice. In other words, the sender’s choices were always deterministic, and we constructed a range of stimulus sets (i.e., sequences of sender’s choices) that varied the stability of the sender’s preference, allowing us to construct scenarios for which the decision function underlying each ABM would identify different rates at which agents converge on their recognition of the sender’s preference. In all cases, the sender’s signal across rounds indicated a dominant choice (i.e., a skew toward red), so that evidence of an underlying preference from the sender despite noise was present in all conditions to varying degrees.

Introducing noise in the observability of the senders’ preference allowed us to control whether participants were exposed to enough signals to accurately infer the sender’s preference according to the TP. We defined four conditions: a 16-round game (target ratio 5:8), a 18-round game (target ratio 6:9), a 20-

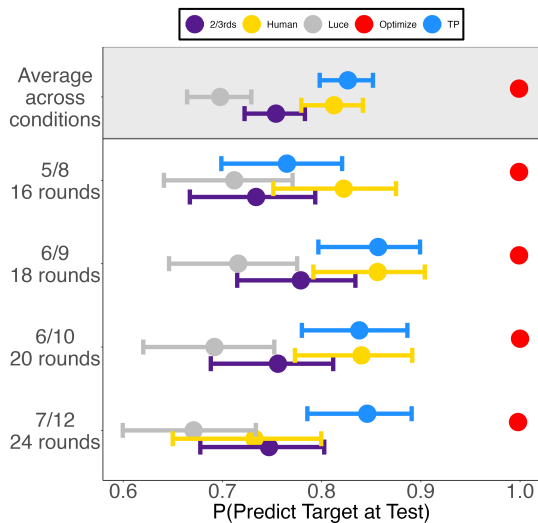


Fig. 5. The fraction of agents correctly identifying the target item in the mind-reading game across all decision models as compared to human participants. Colors indicate the source of data. The shaded gray region shows the results for each model compared to human data aggregated across all conditions. The remaining rows show the by condition breakdown. The predictions for each model were generated by 1,000 simulations of 200 agents for each stimulus set, corresponding to the identical sample size among human participants. In all simulations, the memory size of all agents is drawn from a normal distribution with a mean of 12 and variance of 2. These results hold across a wide range of plausible memory size distributions (*SI Appendix, Table S10*). Error bars show 95% CIs.

round game (target ratio 6:10), and a 24-round game (target ratio of 7:12). Within each condition, we constructed three randomly generated sequences satisfying the corresponding target ratio. Consider, for example, the condition with 20 rounds. The participants are randomly assigned one of three sequences, each of which has the red:total ratio of 6:10. After each round, the participant receives feedback on the color choice they have just made: a small monetary reward is given if it matches the simulated partners' choice, and a small penalty incurs when there is no match with the net amount never allowed to go negative. After all 20 rounds, the participants are instructed to provide a final guess of the sender's true preference for a bonus reward: this is recorded as their decision (*SI Appendix, Fig. S16*). In this design, 200 participants were randomly assigned to each condition (i.e., each round-ratio version of the game), yielding 800 participants total.

To generate model predictions, we used the learning mechanisms of each ABM to simulate the rate at which agents accurately identify the sender's underlying preference in each of the three stimulus sets for each condition (assuming $N = 200$ per condition). We ran 1,000 simulations for each model on each stimulus set in each condition. Each model yielded distinct point-estimate predictions for the probability that agents correctly predict their partner's target choice for each condition, which we preregistered. We then tested each model's ability to predict the actual fraction of participants who successfully identified their partner's coordination preference in the experiment. We hypothesized that TP would provide the most accurate predictions of the experimental data.

Predicting Coordination in Noisy Environments. Fig. 5 shows that, across conditions, the TP model provides the most accurate fit to the fraction of participants who correctly identified the target option in the mind-reading game, as predicted by our

preregistered simulations. TP is the only model for which the fraction of agents identifying the target item is statistically indistinguishable from the experimental data ($P = 0.51$, $CI = [-0.02, 0.05]$, $\chi^2 = 0.42$, Two-sample Proportion Test, Two-sided); all other models yield learning rates significantly different from the human data (all comparisons are significant at the $P < 0.01$ level according to a Two-sample, Two-sided Proportion test). The overall error rates of each model (defined as the absolute difference in convergence rate between a given decision model and the corresponding fraction of human participants selecting "red") are 0.04 (TP), 0.07 (2/3rds), 0.12 (Luce), and 0.19 (Optimize).

TP's superior performance is consistent across conditions. Fig. 5 shows how in three of the conditions, TP achieves the best prediction; it is only in the condition with 24 rounds and a 7/12 ratio that the 2/3rds rule and the Luce model outperform TP, with no significant differences in the accuracy of the Luce model and 2/3rds model ($P = 0.45$, $CI = [-0.13, 0.05]$, $\chi^2 = 0.56$, Two-sample Proportion Test, Two-sided). However, in one of the stimulus sequences in this condition, TP achieves the exact same convergence rate as the human data (87%), which is markedly more accurate than the other models (*SI Appendix, Table S9*); yet, in one sequence, the human convergence data are anomalously low, potentially reflecting sampling noise, which skews the average error of models in this condition. Overall, the success of TP holds across sequences in each condition (*SI Appendix, Table S9*) and when varying i) the memory window size of agents (*SI Appendix, Table S10*) and ii) the memory model (*SI Appendix, Fig. S18*).

Discussion

It is important to stress the scope and limitations of our approach. The TP is a mechanism for implicit learning with child language acquisition as a paradigmatic case; its application to social learning is appropriate for similar tasks such as the name game or the mind-reading game, which, by design, remove extrinsic information such as normative valuation (e.g., signaling ingroup membership or political agreement) or sanctions (such as punishment or social judgment) so as to isolate the cognitive process of rule-based pattern detection underlying conventionalization. Moreover, to stay close to the fundamental game-theoretic assumptions of paradigmatic coordination dynamics, both of the experiments we examine assume there are no intrinsic quality differences in the set of possible coordination behaviors, nor features of the social environment (such as public signals of perceived behavior quality), both of which are established contributors to the evolution of conventions in the wild (12, 80–82). Studying how such factors influence threshold dynamics in convention learning is a valuable direction for future research.

We study the name game because it encompasses conventionalization that takes place endogenously without a centralized authority and where the extrinsic value of options (i.e., their payoffs beyond coordination) are not available, embodying the longstanding puzzle of how conventions spontaneously crystallize from within a state social discord (1). Replicating our findings in the mind-reading game generalizes the predictions of TP to a minimal environment involving the coordination of a nonlinguistic behavior under varying conditions of noise within a general information-theoretic binary signaling task. Together, these controlled studies allow us to isolate and evaluate the basic science question regarding how the cognition of pattern detection supports the learning of conventions (viewed as rule-like patterns

of observed behavioral signals). Identifying threshold dynamics under these general settings helps to form a clear theoretical basis upon which studies of conventions and norms in messier and more complex social environments can build. In future work, our model may also serve as a baseline to measure the extent to which individuals harness extrinsic information to promote coordination; for example, prior work on social contagion demonstrates not only that threshold-based mechanisms can enhance the prediction of the real-world spread of behavior (54), but also that extrinsic information, such as homophily among interactants, informs and influences the threshold dynamics that govern individuals' willingness to adopt new behaviors (53, 83).

A further limitation of our modeling approach to the name game is that we compare ABMs using a relatively small sample of experimental networks. We are reassured that our simulations continue to yield the same findings regarding the outsized performance of TP when we vary network size to examine networks that are much larger than the observed empirical cases. Work on language acquisition further confirms that TP is a robust model of grammar learning even in cases where the set of observations is large—such as linguistic rules that are learned over dozens and potentially hundreds of words encountered over a span of months or years—suggesting that the effectiveness of TP in the name game is not an artifact of the number of observations that participants encounter (84).

That the same simple equation may capture both the individual learning of grammatical rules and the social learning of behavioral conventions has important implications. For one, optimization-based models of human behavior may overlook some of the simple, categorical processes that characterize human social learning as satisficing. Despite their many achievements, the accelerating embrace of AI methodologies (and LLMs specifically) as a framework for simulating human learning is likely to maintain this blind spot, since such models are prevalently designed to emulate human behavior through a brute force, statistically optimized approach. Consistent with recent critiques of such models (85, 86), our findings highlight the benefit of adopting a theoretically driven, psychologically grounded angle toward modeling human social learning, since our framework not only provides more accurate predictions of behavioral data across distinct experimental paradigms, but it also does so with a simple, highly interpretable mathematical model that synthesizes findings across the social sciences (from linguistics, psychology, economics, and sociology).

More generally, demonstrating that it is possible for any simple equation to effectively model a diverse range of learning dynamics is evidence of how ostensibly complex behaviors may be driven by surprisingly simple mechanisms. We anticipate that our results will be of interest to researchers investigating how cognition constrains large-scale social dynamics, such as the emergence and change of new political, economic, and social behaviors (12, 87). Furthermore, we anticipate our framework will be of interest to those investigating how to design AI to better capture the realistic constraints on information processing and rule-based learning that define human cognition and behavior with the goal of building robust simulation frameworks for improving computational social science.

Materials and Methods

The Name Game. The name game was designed by Centola and Baronchelli (6) and replicated by Centola et al. (10). All experimental data from the name

game analyzed in this study was collected by these two studies. The design of the name game is as follows. On each round, each player i is assigned to interact with another player j who is randomly sampled from the set of players that i is connected to according to a network structure. On round t , both i and j independently enter a name for the presented face, which remains constant across all rounds and players. After entering their proposed name on round t , both i and j are shown whether they entered the same name. If the same, both i and j receive a positive payoff [a monetary reward; (6, 10)]; if different, the players receive a minor penalty (a reduction in their total earnings). Both i and j are then randomized to complete this coordination sequence with other players for n subsequent rounds. The authors replicated this design over six trial networks, three of size 24, two of 48, and one of 96, totaling 264 participants. A follow-up study (10) investigated how such conventions may be changed by coordinating defection from a subset of players ("confederates"). After fully connected networks of human players endogenously converged on a shared name, the experimenters seeded a committed minority of confederates who are given instructions to promote an alternative name. Across a total of ten network trials (where networks varied from 20 to 30 people), Centola et al. (10) show that if the size of the committed minority reaches a tipping point (averaging 25% of the network), the established convention flips; but so long as the committed minority remains below the tipping point threshold, the convention remains stable. In total, we examined 16 network trials from Centola and Baronchelli (6) and replicated by Centola et al. (10).

The Mind-Reading Game. The preregistered "the mind-reading game" (<https://osf.io/kqfy5/overview>) was approved by the IRB at Stanford University. This task is designed to compare each learning model (i.e., TP, 2/3rds, OP, and Luce) in their ability to predict the rate at which individuals learn a stable pattern of coordination behavior under varying levels of controlled noise in observed behavioral signals. The task assumes that agents learn by sampling from memory, so the imitation model is disqualified since it assumes agents do not use their memory. The mind-reading game simulates the core pattern learning problem of the name game, yet in the context of coordinating a general, nonlinguistic behavior, namely choices between binary options (in this case, between two colors).

In this task, each player is told they are randomized to play as either a "sender" or a "receiver," where the sender begins by signaling their binary choice, and the receiver then uses the observed signal from the previous rounds to predict the sender's next choice. Yet, by design, all participants were randomized to play the role of receiver, and they were told their task is to predict the choices of their partner who is trying to coordinate with them, since both players receive a reward if the receiving player guesses the sender player's next response. In reality, each participant interacted with a simulated partner whose choices were predetermined. To introduce variation in the stability of the simulated sender's choices, participants were told that a scrambler occasionally misinformed them of the sender's signal, though they were not told how often this scrambler operated. This prompted participants to try to infer the sender's stable underlying strategy across rounds, generalizing beyond the possibly noisy feedback on any given round. However, there was no scrambler. Instead, the stability of the sender's behavioral signal was tuned differently in each condition to derive different predictions from each decision model in terms of the probability that a given agent identifies the sender's preferred color choice. In all cases, the signals received by the receiver indicated a dominant choice (i.e., a skew toward red), so that evidence of an underlying preference from the sender despite noise was present in all conditions to varying degrees. *SI Appendix, Fig. S16* gives a schematic overview.

We randomized 200 participants to one of four classes of sender sequences. Participants played in one and only one of the following conditions: i) 16 rounds with a 5/8 target/total ratio; ii) 18 rounds with a 6/9 target/total ratio; iii) 20 rounds with a 6/10 target/total ratio; and iv) 24 rounds with a 7/12 target/total ratio. For each condition, we selected three distinct round-by-round color sequences (alternating red and blue) satisfying the same ratio (*SI Appendix, Fig. S17*). Participants were only shown one of the color sequences within one

condition. 800 participants in total were recruited; 782 participants finished the task and accurately completed a preregistered attention check satisfying inclusion in our analyses, exhibiting a low attrition rate of 2.25%. To measure whether participants correctly identified the target item, on the final page of the experiment, participants were asked to indicate which color they think the sender was trying to send them all along—blue or red? In all cases, red was the target item. We then determined the fraction of participants selecting “red” and compared this to the fraction of agents selecting red in each decision model. To generate the predictions from each model, we ran 1,000 simulations per model for each condition, assuming $N = 200$. In our preregistration, we provide the exact convergence rates predicted by each decision model in each condition. Specifically, we identified the fraction of agents who reached a deterministic selection of the target item, plus the fraction of agents who did not yet reach a deterministic decision but who still accurately identified the target item probabilistically as a function of the Luce choice axiom, see [SI Appendix, section A.17](#)).

Data, Materials, and Software Availability. Data and code have been deposited in GitHub (<https://github.com/scaplan/name-game>) (88). All other data are included in the manuscript and/or [SI Appendix](#).

ACKNOWLEDGMENTS. We are grateful to Russell Legate-Yang, Amir Goldberg, Dan Zeltzer, Andrea Baronchelli, Damon Centola, Scott Page, Fabian Tschopenig, Eaman Jahani, Ethan Nadler, and Jerry Zhang for helpful feedback. We also thank the audience at the City University of New York Cognitive Science colloquium and the Stanford MACRO Organizational Behavior workshop for useful discussion.

Author affiliations: ^aStanford Graduate School of Business, Stanford University, Palo Alto, CA 94305; ^bLinguistics Program, City University of New York Graduate Center, New York, NY 10016; ^cCognitive Science Program, City University of New York Graduate Center, New York, NY 10016; ^dDepartment of Linguistics, University of Pennsylvania, Philadelphia, PA 19104; and ^eDepartment of Computer and Information Science, University of Pennsylvania, Philadelphia, PA 19104

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